5A - 6A - 10A - Atmosphere - EvolLand

Test 3 Master Test Review

38 Review Questions

TEACHER EDITION

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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TEACHER EDITION

Test 3 Master Test Review

38 review questions in subject order, matching the student edition question for question. Each item gives a model answer and where students find it (linked). The QTI line gives the Canvas bank code and that bank's question number on each past test it appeared on (Fa25 = Fall 2025, Sp26 = Spring 2026; numbers follow the quiz's built order, since Canvas shuffles questions). The JSON line gives the question number in each episode's TeacherCompose file, with its ID. A dash means no match exists.

WHERE TO FIND THE ANSWERS	PASSWORD
Module 5A	Module5A
5AE1	5AE1SG
5AE2	5AE2SG
Atmospheric Chart	Atm
Atmospheric Pressure exercise	AtmosSG
Ancient Atmospheres video	AASG
EvolLand exercise	EvolSG
Module 6A	Module6A
6AE1	6AE1SG
Module 10A	Module10A

The Earth's Atmosphere

1. What is the atmosphere? Excluding water vapor, name the four main gases that make it up, with their percentages.

ANSWER The gaseous envelope that encapsulates the Earth. Nitrogen (N_2) 78%, oxygen (O_2) 21%, argon (Ar) 0.93%, and carbon dioxide (CO_2) 0.04%, about 430 ppm. Ozone and methane are not primary gases.

Find answers in: Module 5A, 5AE1
QTI: T2P22 = Fa25 Test 2 #5, Sp26 Test 2 #5
JSON: 5AE1 Q2 (5AE1-P1Q2); 5AE1 Q3 (5AE1-P1Q3)

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2. Name the four layers of the atmosphere in order, from the ground up, and the "pause" at the top of each. What is a pause?

ANSWER Troposphere (tropopause), stratosphere (stratopause), mesosphere (mesopause), thermosphere (thermopause). A pause is a relatively thin boundary where the temperature trend with height reverses. Memory aid: "Trains Stop More Times."

Find answers in: Module 5A, 5AE1, Atmospheric Chart
QTI: —
JSON: 5AE1 Q5 (5AE1-P1Q5)

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3. In which two layers does the temperature DECREASE with increasing altitude, and in which two does it INCREASE? What heats each of the two warming layers?

ANSWER Decreases: troposphere and mesosphere. Increases: stratosphere and thermosphere. Ozone and oxygen absorbing the Sun's ultraviolet heat the stratosphere; very short-wavelength UV and x-rays from the Sun heat the thermosphere.

Find answers in: Module 5A, 5AE1, Atmospheric Chart
QTI: T2P24 = Fa25 Test 2 #2, Sp26 Test 2 #2
JSON: 5AE1 Q20 (5AE1-P4Q3); 5AE1 Q38 (5AE1-P6Q4); 5AE1 Q43 (5AE1-P6Q9)

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4. In which layer is the ozone layer, and what does it do? Which two gases in that layer protect life from deadly UV radiation? Why is ozone "good up high, bad nearby"?

ANSWER The stratosphere (starting about 15 km, peaking near 25 km). Ozone (O_3) and oxygen (O_2) absorb the most destructive UV wavelengths, which would otherwise damage the DNA of living things. Down in the troposphere, the same ozone is a pollutant that impairs lung function.

Find answers in: Module 5A, 5AE1
QTI: T2P23 = Fa25 Test 2 #4, Sp26 Test 2 #4
JSON: 5AE1 Q21 (5AE1-P4Q4); 5AE1 Q22 (5AE1-P4Q5)

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5. What is the ionosphere, and where is it? Up to what frequency does it reflect radio waves?

ANSWER A region (not a layer) of plasma, electrons and positive ions stripped from atoms by solar radiation, running from about 60 km in the mesosphere to the top of the thermosphere, about 600 km. It reflects radio waves up to about 30 MHz back to the ground.

Find answers in: Module 5A, 5AE1
QTI: T2P27 = Fa25 Test 2 #3, Sp26 Test 2 #3
JSON: 5AE1 Q29 (5AE1-P5Q4)

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6. What sphere harmlessly traps the charged particles of the solar wind?

ANSWER The magnetosphere, Earth's magnetic field.

Find answers in: Module 5A, 5AE1
QTI: T2P35 = Fa25 Test 2 #1, Sp26 Test 2 #1
JSON: 5AE1 Q39 (5AE1-P6Q5)

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7. What is Earth-based weather? Name the six main ingredients of Earth-based weather.

ANSWER Thermodynamic events occurring in the atmosphere (the troposphere), driven by heat moving from warmer to cooler places. The six ingredients: the atmosphere, water (in all three phases), the Sun, the rotation of the Earth, volcanic activity, and biological activity.

Find answers in: Module 5A, 5AE2
QTI: T2P26 = Fa25 Test 2 #6, Sp26 Test 2 #6
JSON: 5AE2 Q2 (5AE2-P1Q2); 5AE2 Q3 (5AE2-P1Q3)

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8. Name the four forms of precipitation, and describe each.

ANSWER Rain: liquid water. Snow: fully formed ice crystals. Sleet: a mix of rain and snow (in the US, often small ice pellets that froze on the way down). Hail: pellets of ice, built up in layers in violent cumulonimbus storms.

Find answers in: Module 5A, 5AE2
QTI: T2P21 = Fa25 Test 2 #7, Sp26 Test 2 #7
JSON: 5AE2 Q7 (5AE2-P2Q2)

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9. Are fog, dew, and clouds forms of precipitation? Explain.

ANSWER No. Precipitation is water falling from the sky. Fog is simply a cloud at ground level, dew condenses directly onto cool surfaces, and cloud droplets stay aloft until they grow heavy enough to fall.

Find answers in: Module 5A, 5AE2
QTI: T2P21 = Fa25 Test 2 #7, Sp26 Test 2 #7
JSON: —

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10. In summer, what are cumulonimbus (storm) clouds primarily composed of?

ANSWER Liquid water droplets (with ice crystals higher up in the cloud).

Find answers in: Module 5A, 5AE2
QTI: T2P28 = Fa25 Test 2 #8, Sp26 Test 2 #8
JSON: —

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11. What is a greenhouse gas? Name three. Why are nitrogen and oxygen NOT greenhouse gases?

ANSWER A gas that lets sunlight in but absorbs part of the infrared heat the ground radiates back out, keeping the surface warmer. Carbon dioxide, methane, water vapor, nitrous oxide, and ozone. Greenhouse molecules have three or more atoms; N_2 and O_2 have only two.

Find answers in: Module 5A, 5AE2
QTI: T2P25 = Fa25 Test 2 #10, Sp26 Test 2 #9
JSON: 5AE2 Q25 (5AE2-P5Q3); 5AE2 Q26 (5AE2-P5Q4)

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12. What makes high and low pressure, and what weather does each bring?

ANSWER Low pressure results from rising warm air; as it cools, its water vapor reaches the dew point and condenses, so lows bring clouds, rain, and snow. High pressure results from sinking cool air, which warms as it is compressed so its clouds evaporate; highs bring clear skies.

Find answers in: Module 5A, 5AE2 (Homework #11)
QTI: T2P30 = Sp26 Test 3 #22
JSON: 5AE2 Q29 (5AE2-P6Q2)

Atmospheric Pressure Exercise

13. Define pressure and atmospheric pressure. How does atmospheric pressure change as you go higher, and why?

ANSWER Pressure is force applied perpendicularly to a surface divided by its area, P = F / A. Atmospheric pressure is the weight of the column of air directly above a given altitude divided by the column's area. It decreases with altitude, because there is less air above you.

Find answers in: Atmospheric Pressure exercise
QTI: —
JSON: ATMOS01-1; ATMOS01-2

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14. Give the value of one Atmosphere in millibars, psi, and inches of mercury (in Hg).

ANSWER 1 Atm = 1,013.25 mb = 14.7 psi = 29.92 in Hg.

Find answers in: Atmospheric Pressure exercise
QTI: T2P31 = Fa25 Test 3 #23, Sp26 Test 3 #20; T2P37 = Fa25 Test 3 #24
JSON: ATMOS01-5

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Ancient Atmospheres

15. What was the Great Oxygenation Event, and which organisms caused it?

ANSWER The start, about 2.4 billion years ago, of a slow rise of oxygen into the atmosphere, which created the first oxygen-ozone shield against UV. It was caused by cyanobacteria, which used sunlight to turn carbon dioxide into oxygen (photosynthesis).

Find answers in: Ancient Atmospheres video (Module 5A, Homework #6)
QTI: T2P39 = Fa25 Test 3 #26
JSON: —

EvolLand Exercise

16. About how many millions of years ago was the Cambrian Explosion, and what happened in it?

ANSWER About 540 million years ago: a rapid burst in the variety of complex, multicellular animal life in the oceans, which set the stage for life to colonize the land.

CHECK PROVISIONAL: drafted without the EvolLand exercise or its guides. Check it against EvolLand, or swap in one of its graded questions.

Find answers in: EvolLand exercise
QTI: T2P38 = Fa25 Test 3 #25
JSON: —

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17. As oxygen in the atmosphere rose, life moved from the sea to the land. About how high did oxygen peak, and when? What happened to flying insects?

ANSWER Oxygen peaked at about 30% around 300 million years ago (it is 21% today). With so much oxygen, flying insects grew to giant sizes.

CHECK PROVISIONAL: drafted without the EvolLand exercise or its guides. Check it against EvolLand, or swap in one of its graded questions.

Find answers in: EvolLand exercise (Module 5A, Homework #6)
QTI: —
JSON: —

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18. In years, about how long ago did our species, Homo sapiens, evolve?

ANSWER About 300,000 years ago.

CHECK PROVISIONAL: drafted without the EvolLand exercise or its guides. Check it against EvolLand, or swap in one of its graded questions.

Find answers in: EvolLand exercise
QTI: T3A32 = Fa25 Test 4 #28
JSON: —

The Nature of Light

19. What is optics?

ANSWER The study of light in all its forms, both visible and invisible.

Find answers in: Module 6A, 6AE1
QTI: T3A2 = Fa25 Test 3 #4, Sp26 Test 3 #3; T3A3 = Fa25 Test 3 #5, Sp26 Test 3 #4
JSON: 6AE1 (NOL) Q1 (NOL-1-01)

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20. What is the simple definition of light?

ANSWER Light is electromagnetic radiation.

Find answers in: Module 6A, 6AE1
QTI: —
JSON: 6AE1 (NOL) Q2 (NOL-1-02)

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21. Rounded, what is the speed of light in a vacuum, in km/s and in miles/s?

ANSWER About 300,000 km/s, or 186,000 miles/s (exactly 299,792,458 m/s).

Find answers in: Module 6A, 6AE1
QTI: T3A9 = Fa25 Test 3 #10, Sp26 Test 3 #9
JSON: 6AE1 (NOL) Q4 (NOL-1-04)

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22. Who first used astronomy to determine the speed of light, when, and how?

ANSWER Ole Rømer, in 1676, by timing Jupiter's moons as they passed behind the planet. When Earth was on the far side of its orbit from Jupiter, the timing ran late because the light had farther to travel. His data led Huygens to about 220,000 km/s.

Find answers in: Module 6A
QTI: T3A7 = Fa25 Test 3 #8, Sp26 Test 3 #7
JSON: —

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23. What does it mean to say laser light is coherent?

ANSWER Coherent light is both temporally coherent (all of one frequency, or monochromatic) and spatially coherent (its wave fronts stay in step, crests lined up, across the width of the beam). Laser light is both.

Find answers in: Module 6A, 6AE1
QTI: T3A5 = Fa25 Test 3 #6, Sp26 Test 3 #5
JSON: 6AE1 (NOL) Q21 (NOL-4-05)

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24. List the electromagnetic spectrum in order of increasing frequency. About what range of wavelengths can humans see? Are cosmic rays part of the spectrum?

ANSWER Radio/TV, microwave, infrared, visible, ultraviolet, x-ray, gamma ray. Visible light runs from about 700 nm (dark red) to 400 nm (dark violet). Cosmic rays are not light; they are high-energy charged particles, mostly protons.

Find answers in: Module 6A, 6AE1
QTI: T3A2 = Fa25 Test 3 #4, Sp26 Test 3 #3
JSON: 6AE1 (NOL) Q23 (NOL-5-02); 6AE1 (NOL) Q26 (NOL-5-05); 6AE1 (NOL) Q28 (NOL-5-07)

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25. What are the three ways light interacts with matter? Define each.

ANSWER Reflection: the bouncing of light off matter. Transmission: the movement of light through something (transparent, translucent, or, if none gets through, opaque). Absorption: light's energy becomes thermal energy in the matter, often re-emitted as lower-frequency light.

Find answers in: Module 6A, 6AE1
QTI: T3A11 = Fa25 Test 3 #2, Sp26 Test 3 #1
JSON: 6AE1 (NOL) Q29 (NOL-6-01); 6AE1 (NOL) Q31 (NOL-6-03)

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26. Name and define the five classical phenomena of light. Which model of light explains all five, and which one can the particle model adequately explain?

ANSWER Reflection (bouncing off matter), refraction (bending as light passes into a medium of different optical density), dispersion (breaking white light into its colors), diffraction (bending around corners), and the Doppler Effect (a frequency shift from relative motion of source and observer). The wave model explains all five; the particle model adequately explains only reflection.

Find answers in: Module 6A, 6AE1
QTI: T3A8 = Fa25 Test 3 #9, Sp26 Test 3 #8; T3A12 = Fa25 Test 3 #12, Sp26 Test 3 #11
JSON: 6AE1 (NOL) Q35 (NOL-7-03); 6AE1 (NOL) Q36 (NOL-7-04); 6AE1 (NOL) Q38 (NOL-7-06)

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27. What is the difference between specular and diffuse reflection? Do all shiny objects exhibit specular reflection?

ANSWER Specular reflection, from smooth, highly reflective surfaces like a mirror, maintains image integrity: you can still recognize the object. Diffuse reflection scatters light in random directions off a surface that is less than shiny and less than smooth, like paper. Every ray still obeys the law of reflection in both. No, not all shiny objects are specular.

Find answers in: Module 6A
QTI: T3A6 = Fa25 Test 3 #7, Sp26 Test 3 #6
JSON: —

The Exploration of Space

28. What was Sputnik 1? Who launched it?

ANSWER The first artificial satellite, launched by the Soviet Union on October 4, 1957.

Find answers in: Module 10A
QTI: T4A7 = Fa25 Test 4 #4, Sp26 Test 3 #25
JSON: —

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29. What was the first satellite the United States successfully launched? What did it discover?

ANSWER Explorer 1 (January 31, 1958). It discovered the Van Allen Belts, zones of charged particles trapped by Earth's magnetic field.

Find answers in: Module 10A
QTI: T4A7 = Fa25 Test 4 #4, Sp26 Test 3 #25
JSON: —

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30. Who was the first person to orbit the Earth?

ANSWER Soviet cosmonaut Yuri Gagarin, aboard Vostok 1 on April 12, 1961.

Find answers in: Module 10A
QTI: T4A2 = Fa25 Test 4 #2, Sp26 Test 3 #23
JSON: —

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31. Who were the first and second Americans in space?

ANSWER Alan Shepard (first, May 5, 1961) and Gus Grissom (second, July 21, 1961), both on suborbital flights.

Find answers in: Module 10A
QTI: T4A6 = Fa25 Test 4 #29, Sp26 Test 3 #24
JSON: —

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32. Who was the first American to orbit the Earth?

ANSWER John Glenn, on February 20, 1962, in Friendship 7.

Find answers in: Module 10A
QTI: T4A2 = Fa25 Test 4 #2, Sp26 Test 3 #23
JSON: —

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33. How many Apollo missions set out to land on the Moon, how many succeeded, and how many men have walked on the Moon?

ANSWER Seven tried to land and six succeeded (Apollo 13 turned back after an explosion). Twelve men walked on the Moon.

Find answers in: Module 10A
QTI: T4A9 = Fa25 Test 4 #30, Sp26 Test 3 #30; T4A10 = Fa25 Test 4 #6
JSON: —

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34. What famous "first" in aviation belongs to Charles Lindbergh? In what plane, and from where to where?

ANSWER The first solo nonstop flight across the Atlantic, in May 1927, in the Spirit of St. Louis, from New York to Paris (about 33 hours).

CHECK Canvas T4A1 asks for "the first person to fly from the United States to Europe" (keyed Lindbergh). Alcock and Brown crossed nonstop in 1919; Lindbergh's first was solo and nonstop. Consider rewording.

Find answers in: Module 10A
QTI: T4A1 = Fa25 Test 4 #1, Sp26 Test 3 #26
JSON: —

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35. Which uncrewed mission produced the first close-up images of Jupiter? What else was first about it?

ANSWER Pioneer 10 (launched 1972). It was also the first spacecraft to cross the asteroid belt.

Find answers in: Module 10A
QTI: T4A3 = Fa25 Test 4 #3, Sp26 Test 3 #27
JSON: —

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36. Which uncrewed mission produced the first close-up images of Saturn?

ANSWER Pioneer 11 (launched 1973).

Find answers in: Module 10A
QTI: T4A3 = Fa25 Test 4 #3, Sp26 Test 3 #27
JSON: —

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37. What were the Viking missions, and what was their key first?

ANSWER Viking 1 and 2 (launched 1975) each sent an orbiter and a lander to Mars. The landers soft-landed in 1976, and Viking Lander 1 sent back the first clear image from the Martian surface.

Find answers in: Module 10A
QTI: —
JSON: —

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38. Which spacecraft is the only one to have visited Uranus and Neptune?

ANSWER Voyager 2 (launched 1977), which also flew past Jupiter and Saturn. It and Voyager 1 are still operating, on very low power.

Find answers in: Module 10A
QTI: —
JSON: —

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